Strain2bScan: strain-level profiling on a 1–2 % genomic subsample

A Database construction (per species)

Reference genomes $G_1 \dots G_n$

digest · 16 type-IIB enzymes

Single-copy 2bRAD tags
25–33 bp · ~1–2 % of the genome
canonical tag → 64-bit marker

single-linkage · Jaccard 0.95
MinHash above 96 genomes

Within-species clusters
(finest unit short reads resolve)

within-species incidence

Marker classes
species-core / shared-partial /
cluster- and strain-specific

unique = absent from every
other cluster at any copy number

cluster × marker database
+ inverted degree index

reference DB

B Sample profiling

Shotgun / WMS reads

in-silico digest

Native BcgI 2bRAD
reads (already tags)

canonical marker counts c_m

the sample is digested ONCE, then matched against every species database

Layer-1 · species gate
present ≥ max(⌈200·r⌉, 10), r = max($1-e^{-\lambda}$, 0.25)

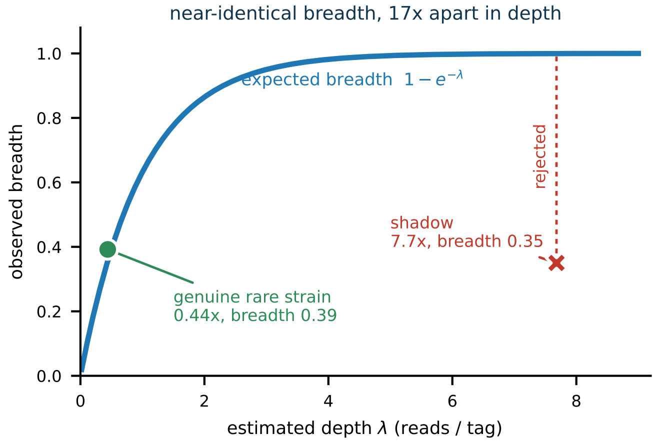
cross-species marker restriction
keep only tags specific to this species panel-wide

Layer-2 · detection
support ≥ 8 · coverage ≥ 0.1
coverage / ($1-e^{-depth}$) ≥ 0.5

depth_j = mean over the WHOLE panel
(zeros included, top 1 % winsorized)

abundance · global_abundance · sample_fraction
within-species · cells · DNA

C What separates a real strain from an artefact



A cluster is called only if its breadth matches
the depth it claims. A strain carrying part of
another cluster's loci fires it at full depth
across a fraction of the panel — and is rejected.

Depth-adaptive evidence
$c \geq 2$ above 3 reads/tag (errors dominate singletons)
$c \geq 1$ below it — at λ=0.5, 78 % of detected
markers are singletons and a fixed rule discards them

Zero-inclusive depth
Averaging only DETECTED markers pins a rare
cluster near 1 read/tag however rare it is,
flattening the whole composition.

Tags are identical to Fast2bRAD-M / 2bRADExtraction.pl → species layer (Fast2bRAD-M) ∘ strain layer (Strain2bScan) share one tag space · Rust, no third-party dependencies · streaming gzip I/O, peak RSS ~11 MB on 4 M reads